

Performance Metrics

CSC3034 Computational Intelligence

Dr Tang Tiong Yew

Introduction

Gold Standard

Percent Correct

Errors

Average Sum-Squared Error

Absolute Error

Normalised Error

Mann-Whitney U Test

Matrices

Contingency Matrices

Confusion Matrices

Receiver Operating Characteristic (ROC) Curves

Chi-Square Test

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- ▶ For example in supervised neural network trainings, the data is provided in the form of input patterns and expected output patterns. The data of expected output patterns is the gold standard.
- ▶ Gold standard is used to compare with the output of a system to evaluate the performance of the system.

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This month, 85 percent of the stocks that outperform were predicted accurately, and 55 percent of the stocks that perform less well were predicted correctly.

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Last month, 90 percent of the stocks that outperform were predicted accurately, and 60 percent of the stocks that perform less well were predicted correctly.

This month, 85 percent of the stocks that outperform were predicted accurately, and 55 percent of the stocks that perform less well were predicted correctly.

The percent correct decreased, however, the overall ability to predict could have improved.

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and 70 percent of the stocks this month outperformed the Dow Jones average,

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The overall predictive accuracy increased, even though the predictive accuracy on the individual metrics decreased.

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- ▶ we might want to consider more than one performance metric in evaluating a system.

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This manipulation of percent correct metric introduces subjectivity, but improve its usefulness in practical sense.

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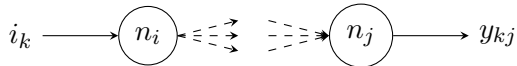
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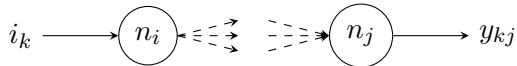
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$$\text{total error, } E_t = 0.5 \sum_{k=1}^m \sum_{j=1}^q (b_{kj} - y_{kj})^2$$

$$\text{average sum-squared error, } e = \frac{E_t}{m}$$

where m is the total number of patterns
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Many implementations ignore this factor to reduce calculation time.

Therefore it is important to be aware of how the error term is calculated when you are using average sum-squared error to compare networks.

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$$e \text{ per output neuron} = \frac{\text{average sum-squared error, } e}{\text{number of output neurons, } q}$$

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Mean absolute error

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Maximum absolute error

$$E_{\max} = \max_{k,j} |b_{kj} - y_{kj}|$$

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$$\text{normalised error, } E_n = \frac{E_t}{E_{\text{mean}}}$$

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Overall it is believed that this error measure may be the most useful one for back-propagation neural network implementations.

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To overcome this situation, it is suggested to remove the output neurons whose expected output does not change value.

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- ▶ Also known as Mann-Whitney-Wilcoxon test or the Wilcoxon rank-sum test.

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- A set of small navigation icons typically found in Beamer presentations, including symbols for back, forward, search, and other slide controls.

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- ▶ The test evaluates whether the medians of two samples of data are the same using hypothesis testing.
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Calculation of the U value:

Method 1 quick and direct

Method 2 formula

Mann-Whitney U test

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n_1 and n_2 are number of runs made with configurations A and B respectively.

$$\therefore n_1 = 5, n_2 = 4$$

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Method 1: quick and direct

1. count the number of A entries that are better than each of the B entries.

	1 st B	2 nd B	3 rd B	4 th B
# of A better than				

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# of A better than	1			

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- count the number of A entries that are better than each of the B entries.

	1 st B	2 nd B	3 rd B	4 th B
# of A better than	1	1	2	

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Method 1: quick and direct

1. count the number of A entries that are better than each of the B entries.

	1 st B	2 nd B	3 rd B	4 th B
# of A better than	1	1	2	5

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$$U_1 = \text{Total} = 9$$

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# of A better than	1	1	2	5	

2. count the number of B entries that are better than each of the A entries.

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# of B better than	0				

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2. count the number of B entries that are better than each of the A entries.

	1 st A	2 nd A	3 rd A	4 th A	5 th A
# of B better than	0	2			

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# of A better than	1	1	2	5	$U_1 = \text{Total} = 9$

2. count the number of B entries that are better than each of the A entries.

	1 st A	2 nd A	3 rd A	4 th A	5 th A
# of B better than	0	2	3		

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2. count the number of B entries that are better than each of the A entries.

	1 st A	2 nd A	3 rd A	4 th A	5 th A	$U_2 = \text{Total} = 11$
# of B better than	0	2	3	3	3	

Next, we arrange the measurements in the order of fitness with better fitness on the left.

0.047	0.049	0.055	0.062	0.069	0.073	0.079	0.085	0.102
A	B	B	A	B	A	A	A	B

Method 1: quick and direct

1. count the number of A entries that are better than each of the B entries.

	1 st B	2 nd B	3 rd B	4 th B	
# of A better than	1	1	2	5	$U_1 = \text{Total} = 9$

2. count the number of B entries that are better than each of the A entries.

	1 st A	2 nd A	3 rd A	4 th A	5 th A	
# of B better than	0	2	3	3	3	$U_2 = \text{Total} = 11$

3. The smaller of U_1 and U_2 is the U value, i.e. 9.

Next, we arrange the measurements in the order of fitness with better fitness on the left.

0.047	0.049	0.055	0.062	0.069	0.073	0.079	0.085	0.102
A	B	B	A	B	A	A	A	B

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	1 st A	2 nd A	3 rd A	4 th A	5 th A	
# of B better than	0	2	3	3	3	$U_2 = \text{Total} = 11$

3. The smaller of U_1 and U_2 is the U value, i.e. 9.

* Note that the two values add up to $n_1 n_2$.

Mann-Whitney U test

0.047	0.049	0.055	0.062	0.069	0.073	0.079	0.085	0.102
A	B	B	A	B	A	A	A	B

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Method 2: formula

Mann-Whitney U test

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1. label the rank from 1 to 9 going from left to right.
2. calculate R_1 , the sum of ranks for A's, and U_1 .

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Method 2: formula

1. label the rank from 1 to 9 going from left to right.
2. calculate R_1 , the sum of ranks for A's, and U_1 .

$$R_1 = 1 + 4 + 6 + 7 + 8 = 26$$

Mann-Whitney U test

0.047	0.049	0.055	0.062	0.069	0.073	0.079	0.085	0.102
A	B	B	A	B	A	A	A	B

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1. label the rank from 1 to 9 going from left to right.
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$$R_1 = 1 + 4 + 6 + 7 + 8 = 26$$

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

Mann-Whitney U test

0.047	0.049	0.055	0.062	0.069	0.073	0.079	0.085	0.102
A	B	B	A	B	A	A	A	B

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$$R_1 = 1 + 4 + 6 + 7 + 8 = 26$$

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1 = 5(4) + \frac{5(5 + 1)}{2} - 26 = 9$$

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A	B	B	A	B	A	A	A	B

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$$R_2 = 2 + 3 + 5 + 9 = 19$$

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4. The smaller of U_1 and U_2 is the U value, i.e. 9.

As mentioned before, there are two steps in a Mann-Whitney U test.

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2. determine if the null hypothesis should be rejected.

Mann-Whitney U test

Our null hypothesis is that the medians of the two samples are equal.

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If the medians of the two samples are equal, the two samples will be evenly distributed across the rank of fitness. Therefore U_1 and U_2 will be similar and at a maximum value.

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For example,

I	II	I	II	I	II	I	II	I	II
---	----	---	----	---	----	---	----	---	----

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I	II	I	II	I	II	I	II	I	II
---	----	---	----	---	----	---	----	---	----

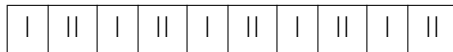
If the medians of the two samples differ a lot, the two samples will be separated clearly on the rank of fitness. U_1 and U_2 will be very different and the U value (the smaller value) will be small.

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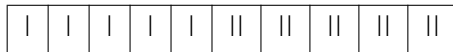
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This is the critical value that we will obtain from the Mann-Whitney U table, below which we will reject the null hypothesis.

Therefore, to determine if the null hypothesis should be rejected, we need to

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1. identify the critical value, and
2. if the U value is less than the critical value, reject the null hypothesis, else, we do not reject the null hypothesis.

The critical values of the Mann-Whitney U for small values of n_1 and n_2 at significance level of 0.05 are given in the following table:

n_1	n_2							
	3	4	5	6	7	8	9	10
3	0	0	1	2	2	3	4	4
4	0	1	2	3	4	5	6	7
5	1	2	4	5	6	8	9	11
6	2	3	5	7	8	10	12	14
7	2	4	6	8	11	13	15	17
8	3	5	8	10	13	15	18	20
9	4	6	9	12	15	18	21	24
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8	3	5	8	10	13	15	18	20
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For $n_1 = 5$ and $n_2 = 4$, the critical value is 2.

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Mann-Whitney U test

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For $n_1 = 5$ and $n_2 = 4$, the critical value is 2.

As $U = 9 > 2$, we cannot reject the null hypothesis, and therefore we cannot say that one configuration produces fitness values that are significantly higher than the other.

Mann-Whitney U test

Let say we test another configuration, which we named as C.

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Let say we test another configuration, which we named as C. We make five runs using configuration C and we obtain the fitness values of 0.060, 0.045, 0.043, 0.038, and 0.034. Compare with the fitness values of the four baseline cases (B), 0.102, 0.069, 0.055, and 0.049, what is the U value? Is the new configuration C statistically better than the baseline configuration B at 0.05 level of significance?

Mann-Whitney U test

Notes:

Mann-Whitney U test

Notes:

- ▶ The previous table is only a partial table of critical values.

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- ▶ When n_1 or n_2 is more than 20, the sampling distribution of U approaches the normal distribution.

Introduction

Gold Standard

Percent Correct

Errors

Average Sum-Squared Error

Absolute Error

Normalised Error

Mann-Whitney U Test

Matrices

Contingency Matrices

Confusion Matrices

Receiver Operating Characteristic (ROC) Curves

Chi-Square Test

Consider a diagnostic or classification system with one output, i.e. the system will classify the input patterns to one of two classes.

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		System Diagnosis	
		Positive	Negative
Gold Standard Diagnosis	Positive	true positive (TP)	false negative (FN)
	Negative	false positive (FP)	true negative (TN)

Contingency Matrix

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Ratios:

Contingency Matrix

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Ratios:

- accuracy

$$\frac{TP + TN}{TP + TN + FP + FN}$$

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Gold Standard Diagnosis	Positive	true positive (TP)	false negative (FN)
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Ratios:

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$$\frac{FP}{(FP + TN)}$$

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$$\frac{TP}{(TP + FP)}$$

- ▶ false positive ratio/false alarm rate

$$\frac{FP}{(FP + TN)}$$

- ▶ true negative ratio/specificity

$$\frac{TN}{(FP + TN)}$$

- ▶ false positive ratio = 1 - specificity

Contingency Matrix

Recall

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Precision (a.k.a. positive predictive value)

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		System outputs		
		A	B	C
Gold standard	A			
	B			
	C			

To construct a confusion matrix, we start by populating the table with the counts of output diagnoses/classes in each cell.

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For example, we consider a medical diagnostic system with three diagnoses A, B and C. A neural network is trained.

Then we use 50 test cases of each diagnosis to obtain the following **confusion matrix**.

		System outputs		
		A	B	C
Gold standard	A	40	8	2
	B	6	42	2
	C	1	1	48

Next we divide each number by the total number of instances in its row to obtain the **class confusion matrix**.

Confusion matrices

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		System outputs		
		A	B	C
Gold standard	A	0.80	0.16	0.04
	B	0.12	0.84	0.04
	C	0.02	0.02	0.96

Next we divide each number by the total number of instances in its row to obtain the **class confusion matrix**.

		System outputs		
		A	B	C
Gold standard	A	0.80	0.16	0.04
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	C	0.02	0.02	0.96

Each row should add up to 1. This is the probability to be diagnosed as a certain class given that the case belongs to a specific diagnose.

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Each row should add up to 1. This is the probability to be diagnosed as a certain class given that the case belongs to a specific diagnose. For example, the numbers in the first row are the probabilities of the case being diagnosed by the system to be A, B, or C given that the case is a case of A.

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The diagonal values represent the recall of each class.

Then we multiply each element by the prior probability of the class of its row to obtain the **final confusion matrix**.

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		System outputs		
		A	B	C
Gold standard	A	0.480	0.096	0.024
	B	0.036	0.252	0.012
	C	0.002	0.002	0.096

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		A	B	C
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Each element is now the probability of that outcome. The sum of all elements is now equal to 1.0.

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$$\text{accuracy} = 0.480 + 0.252 + 0.096 = 0.828$$

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		A	B	C
Gold standard	A	0.480	0.096	0.024
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	C	0.002	0.002	0.096

$$\text{accuracy} = 0.480 + 0.252 + 0.096 = 0.828$$

i.e. 82.8% of the diagnoses the system makes is correct.

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Time to inject some reality. In real life, there are consequences to wrong diagnoses.

In medical diagnoses, the consequences are the treatment cost, human lives, legal cost, etc.

Therefore, we need to construct the cost matrix to investigate the cost of using the system.

Confusion matrices

Suppose that,

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Condition C is a very serious condition that requires hospitalisation and surgery, with a total cost of \$5000. If condition C is not diagnosed right now, the patient has only a 20-percent chance of survival. Even though it is impossible to assign a dollar value to a human life, let's say that the average insurance policy of these patients is \$100,000. That means misdiagnosed condition C will result in an average cost of \$80,000 (80% chance of death $\times$ \$100,000).

It is often assumed that correct classification/diagnosis imposes zero cost, but it is clear that in our case, the cost for a correct diagnosis is not zero.

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Now we are ready to identify the cost for each misdiagnosis and correct diagnosis based on the previous slide.

Cost matrix

Cost matrix

		System outputs		
		A	B	C
Gold standard	A	\$10	\$110	\$5,010
	B	\$110	\$100	\$5,100
	C	\$80,010	\$80,100	\$5,000

Cost matrix

		System outputs		
		A	B	C
Gold standard	A	\$10	\$110	\$5,010
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Then we multiply this cost matrix with the final confusion matrix element by element to obtain the **final cost matrix**.

Final cost matrix

Final cost matrix

		System outputs		
		A	B	C
Gold standard	A	4.8	10.56	120.24
	B	3.96	25.2	61.2
	C	160.02	160.20	480.00

Final cost matrix

		System outputs		
		A	B	C
Gold standard	A	4.8	10.56	120.24
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The average estimated cost of each application of the diagnostic system is given by the sum of all elements in the final cost matrix, which is \$1,026.18.

The cost calculations from confusion matrices can be used to fine-tune a computational intelligence system.

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For example a neural network-based diagnostic system is first trained. Then the threshold levels of the output neurons can be adjusted by calculating the costs associated to choose the combination that produces the lowest system cost.

The cost calculations from confusion matrices can be used to fine-tune a computational intelligence system.

For example a neural network-based diagnostic system is first trained. Then the threshold levels of the output neurons can be adjusted by calculating the costs associated to choose the combination that produces the lowest system cost.

Even if cost information is not available, the system can still be tuned to minimise the off-diagonal numbers, of which the sum represents the percent incorrect.

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Receiver Operating Characteristic (ROC) Curves

Chi-Square Test

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Receiver operating characteristic (ROC) curves

- ▶ Receiver operating characteristic (ROC) curves are also called relative operating characteristic curves.
- ▶ ROC curves started to be used in 1940s for electronic communication systems and in the field of psychology.

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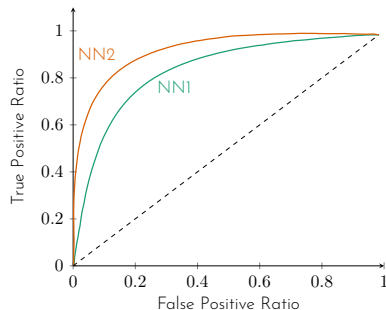
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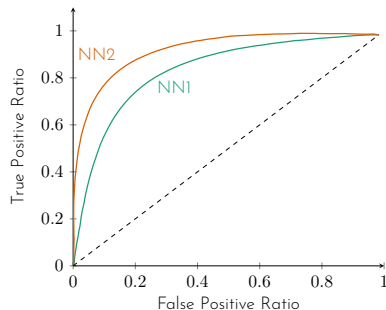
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- ▶ An ROC curve is a plot of the true positive ratio (sensitivity) versus the false positive ratio ($1 - \text{specificity}$).
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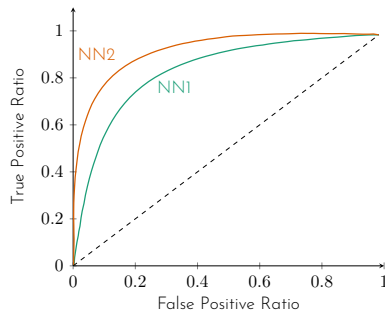


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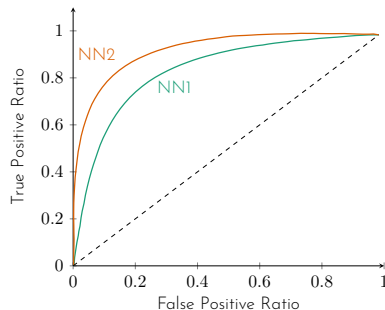
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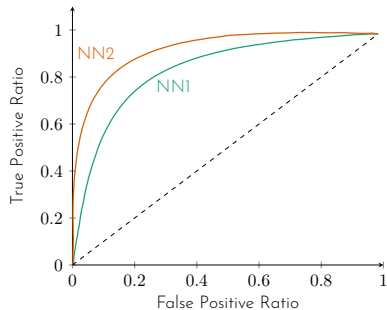
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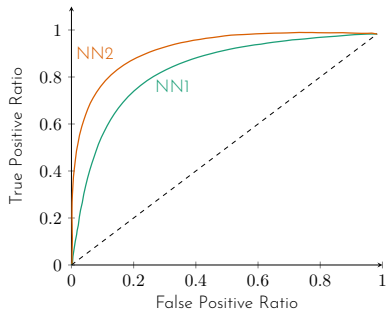
The questions of concern are:

- ▶ How do we interpret the ROC curves?
- ▶ Which is the neural network with better performance?

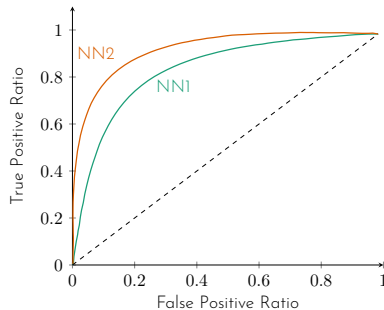
ROC curves



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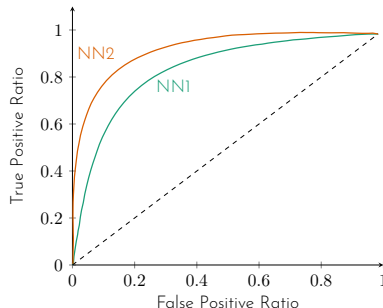


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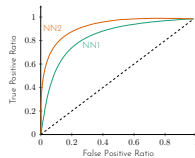
- It always lies above the major diagonal.

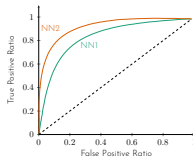


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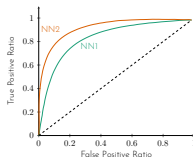
- ▶ It always lies above the major diagonal.
- ▶ It is always monotonically increasing in value from left to right.

ROC curves

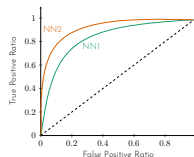




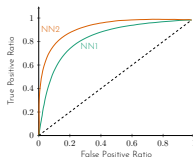
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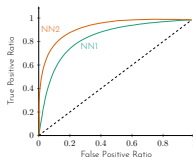
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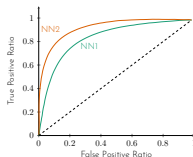
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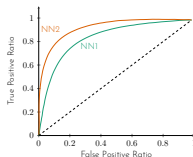
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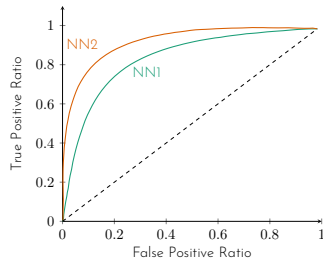


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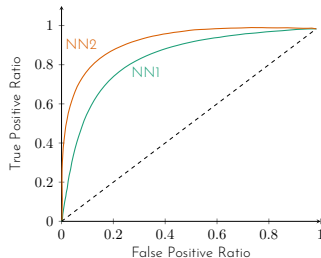


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 - generate a smooth curve through the points and calculate the area under it, or
 - join the points with straight lines and calculate the area using trapezoidal rule.

ROC curves



ROC curves



The shape of the curve indicates the sensitivity of the system performance to the threshold value.

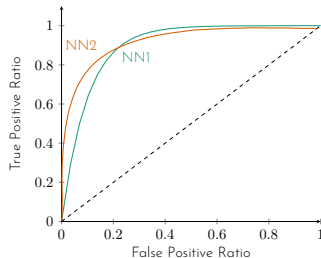
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To use chi-square test, the categories must be **mutually exclusive** and **collectively exhaustive**. The data is assumed to be normally distributed.

Chi-square test

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$$\chi_2^2 = \frac{(2 - 5)^2}{5} + \frac{(15 - 10)^2}{10} + \frac{(8 - 15)^2}{15} + \frac{(25 - 20)^2}{20} = 8.82$$

Chi-square test

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In our example, there are 4 categories, and therefore

$$df = 4 - 1 = 3$$

To obtain critical values from the chi-square distribution table

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df	Area to the right									
	0.995	0.990	0.975	0.950	0.900	0.100	0.050	0.025	0.010	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
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For degree of freedom $df = 3$, at 0.05 level of significance, our critical value is 7.815.

Chi-square test

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The boundary or threshold value for χ^2 for us to conclude that the difference between the two distributions is significant is the critical value.

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Therefore,

If the χ^2 value is greater than or equal to the critical value,

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whereas if the χ^2 value is less than the critical value,

we **do not reject** the null hypothesis.

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$$\chi_2^2 = 8.82 \geq 7.815,$$

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This means that the difference between the expected frequency distribution and the observed frequency distribution in the first configuration is not significant.

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This means that the expected frequency distribution and the observed frequency distribution in the second configuration are significantly different.

Therefore, we would say that the first system configuration is better than the second one.

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Or at the very least, the first system configuration is acceptable while the second one is not.

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Notes:

- ▶ Chi-square test evaluates the performance of the entire system at once, as oppose to ROC curve, which is designed to analyse one output neuron (one category) at a time.
- ▶ It is necessary to determine the categories that are possible. In some cases, not all categories are expected. The categories with zero expected frequency have to be handled with care, whether to remove them, or to include them and perform division by zero.

Chi-square test

Notes:

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- ▶ In general, chi-square test should only be used when $\leq 20\%$ of the expected counts are < 5 , and all individual counts are at least 1.

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- ▶ In general, chi-square test should only be used when $\leq 20\%$ of the expected counts are < 5 , and all individual counts are at least 1.
- ▶ It should be obvious that when designing or tuning a computational intelligence system, the goal is to minimise the chi-square value, i.e. the observed frequency distribution should be as similar to the expected frequency distribution as possible.

A classification system categorises objects into 3 categories, A, B, and C. A test set consists of 20 items for each class is used to perform testing on two classification algorithms. The result of algorithm I shows 30 A, 15 B, and 15 C whereas algorithm II shows 23 A, 18 B, and 19 C. Use χ^2 test to evaluate the two algorithms.

Introduction

Gold Standard

Percent Correct

Errors

Average Sum-Squared Error

Absolute Error

Normalised Error

Mann-Whitney U Test

Matrices

Contingency Matrices

Confusion Matrices

Receiver Operating Characteristic (ROC) Curves

Chi-Square Test

A blue-tinted photograph of a Sunway University building. In the foreground, a large, curved banner is suspended, featuring the Sunway University logo and the text "SUNWAY UNIVERSITY". The banner is supported by several vertical poles. In the background, a multi-story building with a modern architectural style is visible, featuring large windows and a glass facade. The scene is set outdoors, with a paved area in the foreground and some greenery on the right side.

SUNWAY
UNIVERSITY

Thank you

CSC3034 Current Course Update

10 Performance Metrics - presentation

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current labs connect core CI algorithms with physical-AI simulation examples.
- Isaac Sim examples use its bundled Python 3.12 environment and current launchers.
- Every simulator-dependent example includes a CPU/Matplotlib learning fallback.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.